TOPIC 2 Systems of Linear Equations and Inequalities

Name _____ Date _____

Reminders
Boundary: solid for ≤ or ≥, dashed for < or >.
Test (0, 0): true → shade its side; false → shade the other side.
Words: at least ≥ · at most ≤ · more than > · less than <.

1. A \$30 gift card buys songs for \$1.50 (s) and movies for \$6 (m). Which inequality shows what Kim can buy?

> **Tip:** Match each price with its item. "At most" → ≤.

A. $1.5s + 6m \geq 30$

B. $6s + 1.5m \leq 30$

C. $1.5s + 6m \leq 30$

2. Can Kim buy 8 songs and 3 movies with the card?

> **Tip:** Substitute s = 8, m = 3.

F. Yes, it costs exactly \$30

G. No, it costs \$33

H. Yes, it costs \$24

3. A solid line passes through (0, 2) and (4, 0). The half-plane below it is shaded. Write the inequality.

> **Tip:** Find the slope and y-intercept, then pick the symbol.

Slope = $-2 \div 4$ = _____

y-intercept = _____

Solid + below: y ____ _____